Inorganic and Materials Cumulative Exam
Feb 3 2017

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded.

1. Why are quartz cuvettes preferred to plastic or glass ones in a measurement of UV and visible light absorption?

2. Using valence molecular orbital diagrams to justify your answer, why does hexagonal boronitride (hBN) have a band gap while graphene does not?

3. (15pts) Describe the process of multi-exciton generation in quantum dots.
   (5pts) How might this process improve the efficiency of a solar cell comprised of quantum dot

4. Compare and contrast the following techniques: Raman spectroscopy and Electron-Energy-Loss Spectroscopy (EELS). Include basic descriptions of the 'probe' (electrons, neutrons, x-rays, light, etc.) used to measure the spectra, the instrumentation required, and the phenomena measured by these techniques.

5. Provide an example of a molecule that forms an organic semiconductor in solid phase. What property of the molecule determines its bandgap?

6. A buried iron pipe is often connected by a wire to a Mg or Zn rod. Why is this?

7. In an XPS measurement, the Si(2p) peak has a higher binding energy for SiO2 (103.5eV) compared to elemental Si (99eV). Why is this the case? Please explain this chemical shift in qualitative terms with respect to the local electronic environment around the Si atoms in each material.

8. (15 pts) Describe the basic principle behind the operation of a semiconductor LED.
   (5pts) Is it preferable to use a direct gap or indirect gap semiconductor?

9. Why are hydrated Mn2+ ions colorless even though they have 5 3d electrons? Use a molecular orbital diagram to support your explanation.

10. An enhancement in Raman absorption signal is observed for molecules on a rough gold surface. Please provide an explanation of how this surface enhanced Raman spectroscopy (SERS) works (15pts) and why a rough metal surface is required (5pts).